

```
%oldchk=/groups/rmaier/Ravindu/Corrections/PbRhaC14C14C6T1V2/PbRhaC14C14.chk
%nprocshared=64
%mem=800GB
%chk=PbRhaC14C14T1.chk
# opt=(loose,restart) freq apfd/gen scrf=(solvent=water) guess=read pseudo=read
scf=xqc
```

PbRhaC14C14

0 1

O	1.65631563	-2.26584961	-0.11021457
O	4.14358141	-2.52362574	-1.79243506
O	6.76865544	-1.17719652	-0.03139593
O	3.46570970	-4.14604954	0.84038060
C	0.59269620	-6.27002535	-0.39360777
O	-0.38942841	-5.28169016	-0.08606953
C	0.02618403	-4.00286514	-0.51519601
C	1.23897100	-3.55388422	0.28616463
C	2.35976114	-4.52951959	0.03448617
C	1.90286940	-5.95125298	0.34722890
O	6.20598191	-3.25616866	0.46606816
O	3.65030587	-0.37347705	-1.68536995
C	3.67620337	-1.46287028	-2.31064696
C	3.15349033	-1.48948722	-3.73133633
C	1.78494278	-0.83952006	-3.87103654
O	0.77923342	-1.83766215	-3.53725322
C	1.49963987	-0.35829014	-5.28307164
C	0.12668700	0.28033892	-5.43675640
C	-0.12691334	0.80332184	-6.84412521
C	-1.50707462	1.42631828	-7.00429181
C	-1.77229861	1.95315965	-8.40819886
C	-3.15419153	2.57365879	-8.56528988
C	-3.42254635	3.10291397	-9.96771147
C	-0.16839675	-1.53862673	-2.64820736
O	-0.16189331	-0.56737686	-1.92504202
C	-1.27072704	-2.56863752	-2.68730775
C	-1.91800920	-2.82505471	-1.33088442
O	-0.95344784	-3.05858461	-0.28920925
C	0.02912857	-7.61558644	-0.00950673
O	2.93324992	-6.88265456	0.07242702
C	2.72694120	0.76389563	5.38399648
C	3.82013313	-0.30009598	5.19014483
C	3.21008260	-1.58560433	4.62135275
O	1.88793038	-1.74375600	5.09001506
C	0.99775546	-0.79592431	4.48347197
C	1.67082711	0.58184833	4.29644055
C	0.41259189	-1.33742578	3.19088768
O	0.71152602	1.62032159	4.39693416
O	3.97936724	-2.66692451	5.01130458
O	3.28911761	2.05807853	5.40141627

O	4.76816777	0.25086051	4.29943618
C	4.15615539	-3.71635581	4.05024360
C	5.64441628	-4.00165594	3.97323238
C	3.35040247	-4.96166810	4.38359365
C	6.40154899	-2.88673370	3.31030862
O	7.67608447	-3.18957340	3.14889050
O	5.89342914	-1.83666687	2.95527762
C	8.58993703	-2.31786193	2.40658288
C	1.86212694	-4.79862386	4.11532794
C	1.09026672	-6.10740153	4.18503237
C	-0.33711914	-5.97988935	3.66920105
C	-1.09534424	-7.29968746	3.69719161
C	-2.44823416	-7.24099344	3.00185613
C	-3.13757119	-8.59695789	2.93154557
C	8.45998083	-2.61124449	0.90602278
C	7.06452201	-2.33190837	0.38814064
C	8.51350693	-0.83873880	2.76307268
C	8.42840058	-0.55654120	4.25832052
C	8.05996967	0.89287978	4.55309556
C	7.61669031	1.10821743	5.99307636
C	7.10485475	2.51664360	6.25793952
C	6.52637270	2.69095677	7.65528625
C	5.96732464	4.08452659	7.90664216
C	-2.95577518	-3.93302049	-1.41278367
C	-3.72264194	-4.12645846	-0.11315280
C	-4.84192510	-5.15074288	-0.23780955
C	-5.69160305	-5.26442668	1.02046145
C	-6.84438876	-6.24815908	0.87638850
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H	1.72106572	-6.03242064	1.42225063
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H	3.13595534	-2.50964961	-4.11805319
H	1.69161286	-0.01840765	-3.15954757
H	2.27661907	0.36913581	-5.54412615
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H	-1.62258093	2.24428905	-6.28080728
H	-1.00749108	2.69701557	-8.66837710
H	-1.65598820	1.13535842	-9.13184279
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H	-2.65940727	3.84863972	-10.22824534
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H	-0.90989028	-3.48300822	-3.16131258
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H	-0.89574267	-7.81144478	-0.55528820
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H	2.24567158	0.61670976	6.35530191
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H	0.20396978	-0.65187010	5.22191448
H	2.16831648	0.63654008	3.31791569
H	-0.31086140	-0.63224594	2.77194768
H	-0.10140868	-2.28413028	3.36713965
H	1.19185896	-1.49533877	2.44148948
H	0.20656971	1.65191287	3.57916413
H	4.03005543	2.03228929	4.77963684
H	5.26267798	-0.47304413	3.87800499
H	3.82194225	-3.34781343	3.07657935
H	6.06037639	-4.16666832	4.97267388
H	5.83907767	-4.91335641	3.40206019
H	3.73320694	-5.77599980	3.75694825
H	3.53912160	-5.25403631	5.42338957
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H	1.61317156	-6.87381615	3.59564506
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H	-0.87893683	-5.22536086	4.25415538
H	-0.31761929	-5.59972665	2.63908221
H	-0.48093414	-8.07579369	3.22046896
H	-1.22749489	-7.62927704	4.73599229
H	-3.09817750	-6.52213634	3.51798020
H	-2.31685949	-6.84854595	1.98495404
H	-3.28639181	-8.98724445	3.94736024
H	-2.47456087	-9.31345051	2.42766929
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H	8.70083506	-3.66017889	0.72539978
H	9.42088365	-0.38494396	2.34736805
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H	7.66565343	-1.19669389	4.71159526
H	9.37141108	-0.82150543	4.75168446
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H	7.23495489	1.18767879	3.89378249
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H	8.43975796	0.87487567	6.68155494
H	7.91292683	3.24320993	6.09960045
H	6.33212720	2.75863575	5.51675179
H	5.73181574	1.94880446	7.80986794
H	7.29871970	2.46619926	8.40309782

H	5.20186877	4.30963355	7.15195613
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H	-3.65367818	-3.67567543	-2.21879090
H	-2.46581393	-4.86770287	-1.70831380
H	-3.02873196	-4.42639864	0.67891548
H	-4.14930244	-3.16355333	0.19781058
H	-5.48711206	-4.88179041	-1.08498029
H	-4.41608064	-6.13278305	-0.48311863
H	-5.05601956	-5.56431924	1.86323564
H	-6.08935485	-4.27407559	1.27956736
H	-7.48502798	-5.93762653	0.04022713
H	-6.44904564	-7.23364663	0.60042194
H	-7.04194947	-6.63428322	2.98594370
H	-8.13729187	-5.40906357	2.38146981
H	-8.33318361	-8.39543133	1.74603491
H	-9.44932354	-7.16993294	1.17734375
H	4.21671819	-4.71606624	0.63093681
H	0.87808378	-1.69251840	-0.15622831
Pb	4.37790070	-1.60207769	0.37769502
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H	9.55489929	-2.70896663	2.73570902
C	-4.80563037	3.72249989	-10.12439749
H	-5.56660901	2.97608601	-9.86392326
H	-4.92044227	4.53658277	-9.39765688
C	-5.06152215	4.24880513	-11.53109720
H	-4.33121736	5.01799445	-11.80184099
H	-4.98203175	3.44643567	-12.27150222
C	5.36659494	4.25236968	9.29656407
H	6.13165779	4.02727258	10.05031562
H	4.57445631	3.50638682	9.43849254
C	4.80604461	5.64914298	9.53314018
H	5.58626845	6.40935997	9.42500932
H	4.01786719	5.88353060	8.81053891
C	-9.61154240	-7.59590069	3.28158306
H	-10.05920298	-6.63059782	3.54962105
H	-8.94548820	-7.86073814	4.11246943
C	-10.70347643	-8.64912136	3.14201733
H	-10.29103340	-9.63383144	2.90487439
H	-11.40621838	-8.38288536	2.34601575
C	-4.47308176	-8.54844557	2.20432025
H	-5.12712567	-7.82740489	2.70613550
H	-4.31353856	-8.15543045	1.19287643
C	-5.16874210	-9.90036911	2.12909651
H	-5.36904443	-10.29683015	3.12955226
H	-4.55003225	-10.63423090	1.60259953
C	-6.51452623	-9.80650491	1.38636197
H	-7.15869257	-9.12398396	1.90028306
H	-6.34776956	-9.45716212	0.38883913
C	-7.17319680	-11.19775303	1.33965265
H	-7.33780957	-11.54811052	2.33717572

H	-8.10935313	-11.13208453	0.82564976
H	-6.53007520	-11.87971005	0.82367860
C	-11.51263539	-8.78308061	4.44544232
H	-10.85947251	-9.07028843	5.24280668
H	-11.96986876	-7.84442715	4.67946654
C	-12.60378606	-9.85463451	4.26443830
H	-12.14461313	-10.80771677	4.10414345
H	-13.21286108	-9.89545097	5.14322248
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C	-6.46463291	4.86951136	-11.66377717
H	-7.20461650	4.12892852	-11.44273119
H	-6.56018516	5.68506531	-10.97774399
C	-6.66519614	5.38231505	-13.10197283
H	-7.64223451	5.80835064	-13.19572613
H	-5.92882216	6.12707656	-13.32102458
H	-6.56388404	4.56781643	-13.78843321
C	4.20495121	5.79410327	10.94355616
H	4.96331241	5.60861811	11.67525523
H	3.41018383	5.08837447	11.06681070
C	3.65625569	7.22174521	11.12347754
H	3.23781620	7.32219757	12.10312937
H	4.45119763	7.92743017	11.00110088
H	2.89851686	7.40754052	10.39121258

C H O 0
6-311+g(2d,p)

Pb	0	
S	4	1.00
	591.61124370	0.22126521076D-03
	46.757232559	0.56961959130D-02
	20.746462696	-0.21374063831
	14.610796419	0.40502620616
S	2	1.00
	20.222637612	-0.83541883299D-01
	6.4767324865	0.97910892388
S	1	1.00
	1.6600600927	1.0000000
S	1	1.00
	0.80431655001	1.0000000
S	1	1.00
	0.22627039020	1.0000000
S	1	1.00
	0.84014530665D-01	1.0000000
P	3	1.00
	15.189102118	0.61952303583
	14.693144415	-0.72498497086
	6.8705890048	0.37680007984
P	3	1.00
	2.2021426123	0.40196284806

	1.2209125119	0.46058131862
	0.63367559815	0.19367655397
P	1 1.00	
	0.28202837058	1.0000000
P	1 1.00	
	0.11333375666	1.0000000
P	1 1.00	
	0.43948707430D-01	1.0000000
D	6 1.00	
	61.315369628	0.33870800787D-03
	12.372195840	0.13788683942D-01
	6.9254944983	-0.75979608103D-01
	2.3319539939	0.28113784298
	1.2108730003	0.44474512269
	0.60090478506	0.35326874351
D	1 1.00	
	0.28135869813	1.0000000
D	1 1.00	
	0.11500000000	1.0000000
F	1 1.00	
	1.0000000	1.0000000
F	1 1.00	
	0.2896200	1.0000000

PB 0

PB-ECP 3 60

f potential

2

2	3.88751200	12.20989200
2	3.81196300	16.19029100

s-f potential

4

2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

p-f potential

6

2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

d-f potential

6

2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600

2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100